Ethical & Legal Considerations

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***** EVERYTHING HIGHLIGHTED IN GREEN AND UNDERLINED WAS COVERED IN SECOND SPRINT.*****

Legal Considerations

A yearly review is done to ensure that the website follows current standards set by the Government of the United Kingdom namely, GDPR, DPA & WCAG 2.2. The standards are upheld to the best of our capabilities.

1. Privacy Policy (GDPR & DPA)

According to UK General Data Protection Regulation and Data Protection Act 2018, personal data must be [1][7]:

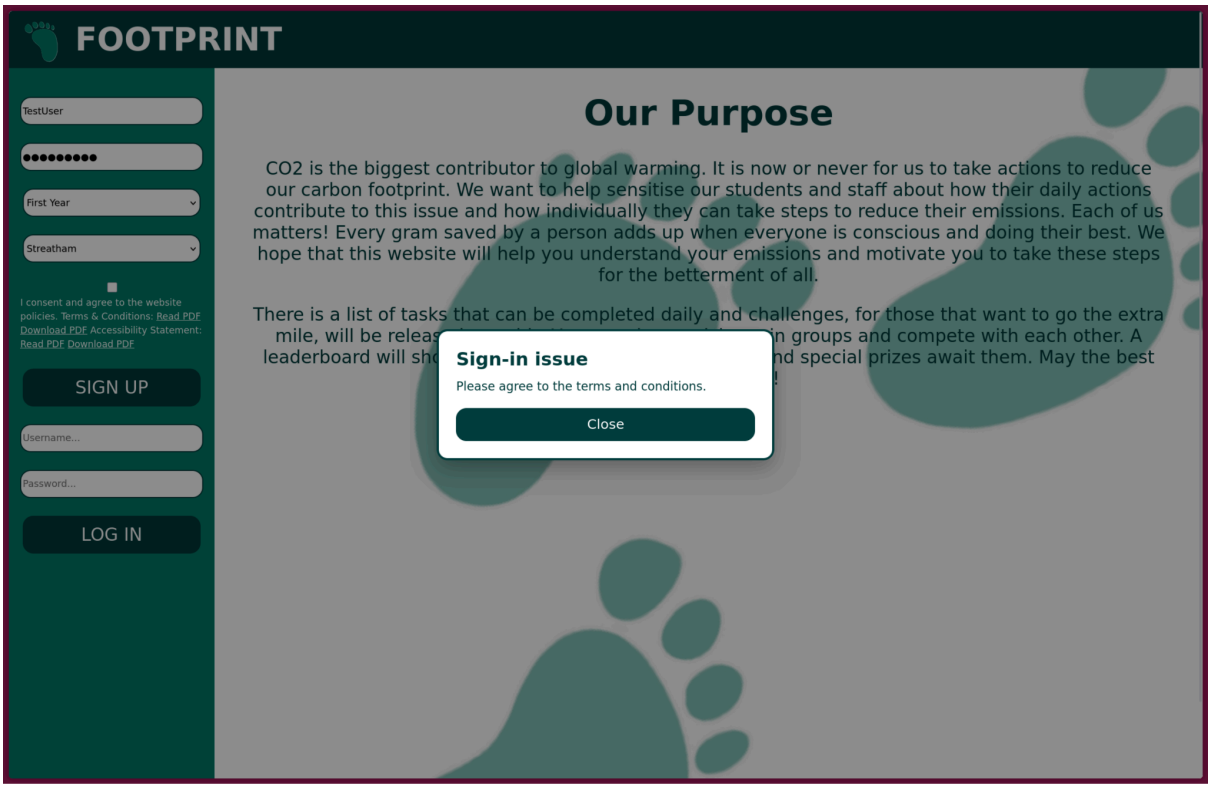
- used fairly, lawfully and transparently
- used for specified, explicit purposes
- used in a way that is adequate, relevant and limited to only what is necessary
- accurate and, where necessary, kept up to date
- kept for no longer than is necessary
- handled in a way that ensures appropriate security, including protection against unlawful or unauthorised processing, access, loss, destruction or damage

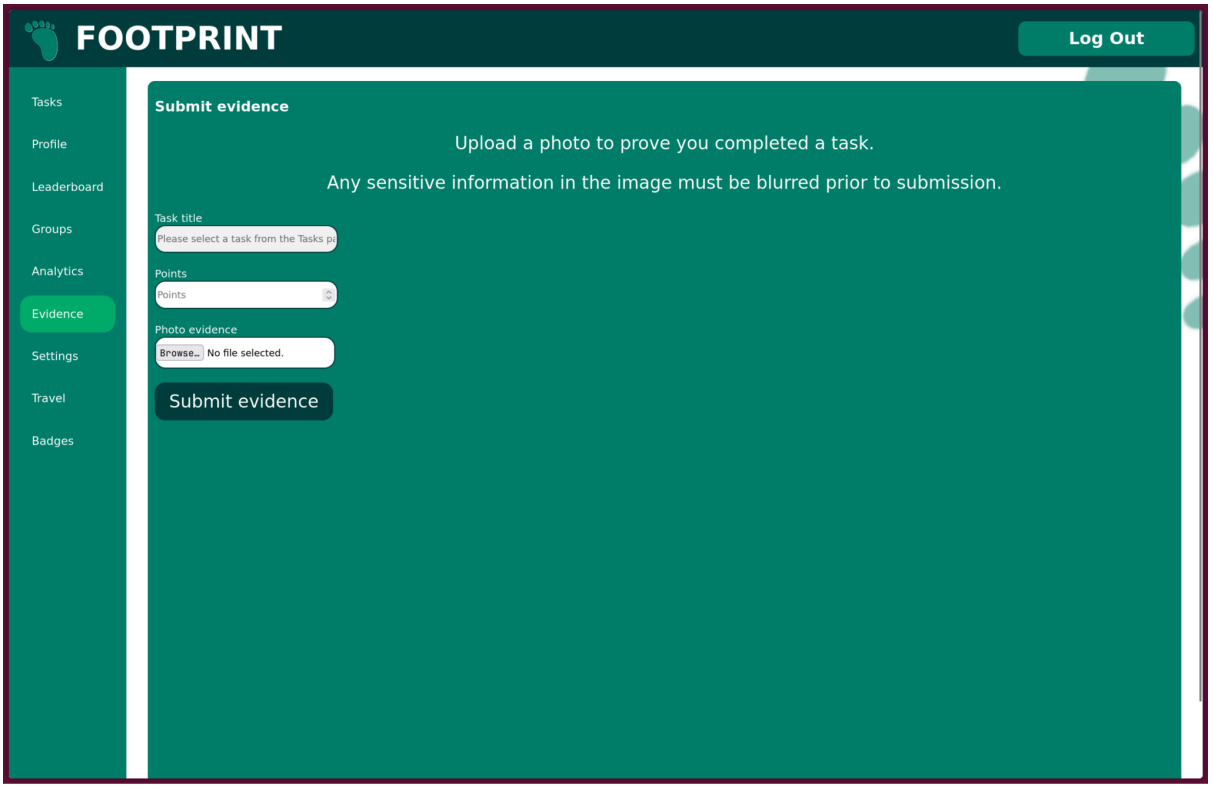
To meet these laws and regulations, we have enforced the following in our system:

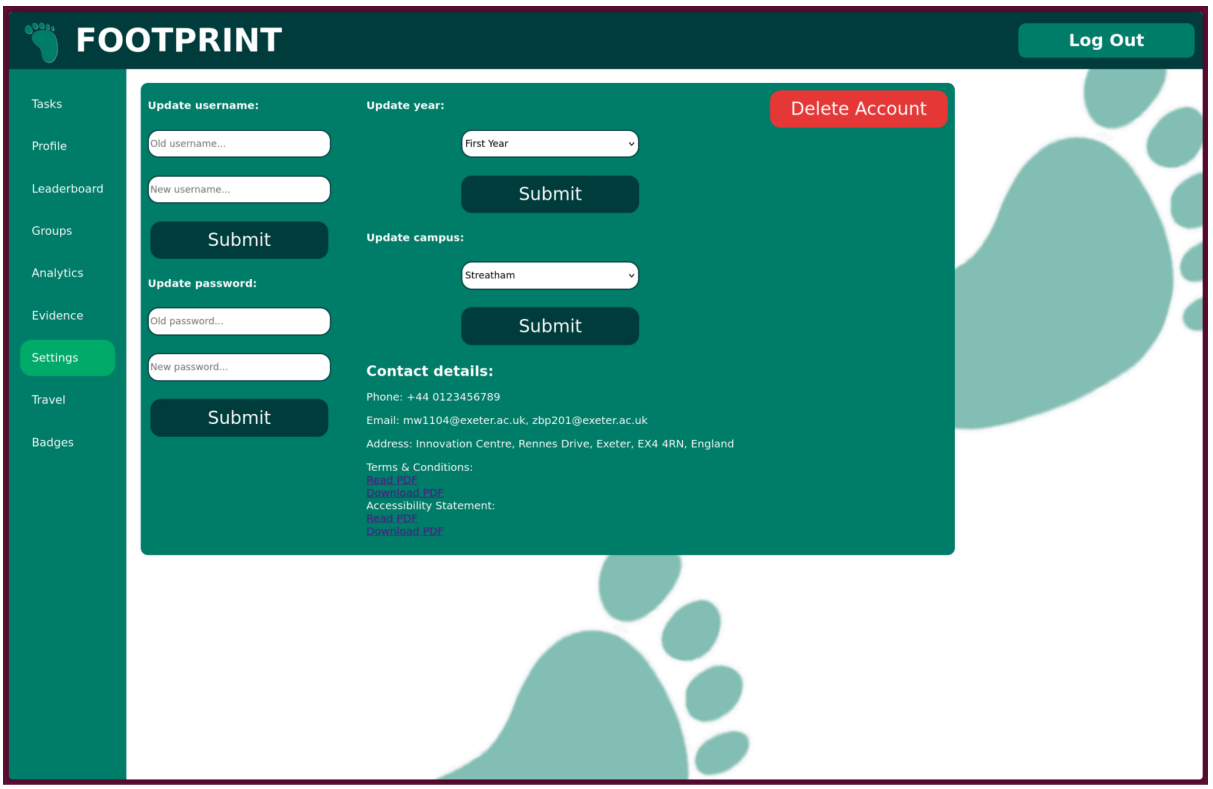
Except for use on the website, personal data is kept private and the user consents beforehand. Furthermore, it is only used for analytics which is clearly stated in the terms and conditions linked on the website and user is required to agree to that to be able to use the website. No sensitive data is needed from the user, only a username and password. Moreover, the user is asked to blur sensitive parts of images submitted for evidence or the evidence is not approved and if the provided evidences are continuously disturbing, the moderators will flag the account to the developers for the latter to delete. The user has the option to change their username, role, password and campus at their will. Images used for evidence are manually deleted by developers after a year [11], considering moderators no longer need them to check and see if the same images are being submitted. Additionally, they also have the option to delete their account and have all their information stored erased. In the same token, all passwords are hashed and the hashed versions are stored instead of plaintext. The *BCrypt password encoder bean* has been used and is known to be safe at the moment of development [3][4]. Also, evidence uploads are kept private as data is called via user ID so users can only retrieve their own information and the latter is securely stored in the database. The service being used to host the website is Render and it encrypts all information(images, username, password, role, campus) both in transit and at rest. In particular, images are stored as raw byte code which gets encrypted

by render as well. Moreover, all the SQL used for input are parameterised thereby preventing SQL injection. The password is also hidden when the user is typing it to prevent its accessibility from onlookers.

The above is demonstrated as shown below:







2. Accessibility Standards

The 4 design principles put forward by the government of the UK need to be followed as far as possible. [5]

- **Principle 1: Perceivable**

  Text alternatives are provided for images to ensure that screen readers can work. Content is structured in a logical order thereby making navigation easier. Moreover, clickable items change colour when you hover on them. Proper markup is used for every feature so that relationships between content are defined properly. Furthermore, the Verdana font is used everywhere on the website as it is more dyslexic-friendly and there is no audio content to keep it simple for users with hearing disabilities. Additionally, text colours are white against a very dark background to ensure readability. Lastly, the website can be zoomed thereby making it more friendly for people affected by low-vision.

- **Principle 2: Operable**

  There is no use of flashing/blinking content to ensure the safety of people who are affected by photo-sensitive seizures . Additionally, descriptive titles are provided for all tabs and users can move through them easily as they are shown at all times and once on the description, the colour will change indicating that it can be clicked and the page is accessed with a click.

Meaningful headings and labels were used giving the users an idea of the following content on the website. Besides, buttons are big enough to be visible and are spaced far enough to prevent accidentally clicking on the wrong button.

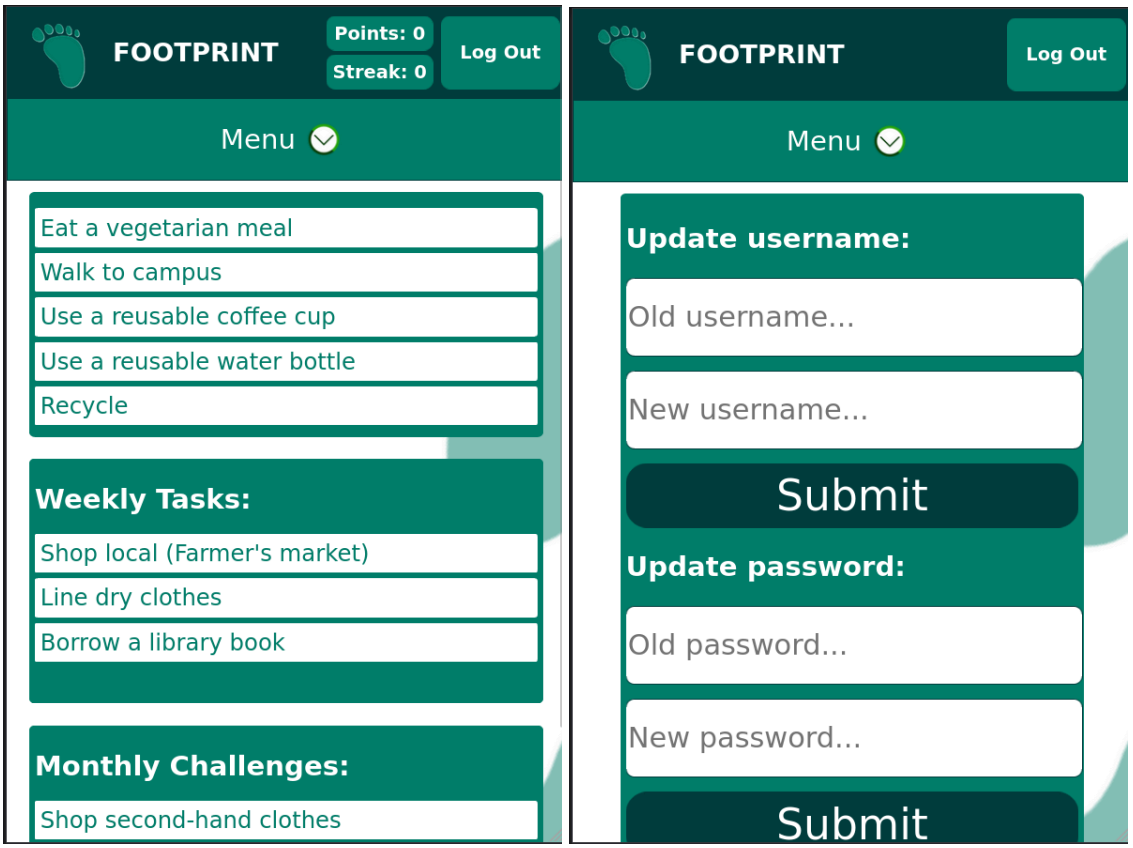
- **Principle 3: Understandable**

  Since the website is created for the University of Exeter, and will be launched by them, the website is in their primary language (English). All features of the website look consistent and a similar colour scheme is adopted across all of them. Furthermore, all form fields follow the same structure and have meaningful and visible labels. Also, log-in is clear-cut and made easy as the user will only have to remember their username and password and no extra action is required of them.

- **Principle 4: Robust**

  The website is coded in clear html thereby making it easy for assistive technologies like screen readers to read the features of the website and also for the website to work on different browsers. In addition to that, all images have an alternative text. Moreover, the website is mobile-friendly, hence all the features are easily accessible on mobile and it is clear on how to use them.

The above is demonstrated as shown below:



3. Terms and conditions [6]

User Agreement
By using our website, footprint, you are agreeing to all the terms and conditions henceforth stated.

Intellectual Property
All text, images and analytics belong to footprint and cannot be used elsewhere without clear permission from developers.

Acceptable Use
You must use the website responsibly and do not spam submissions for challenges. Furthermore, group names must be appropriate, descriptive and not contain any slurs or inappropriate language.

User Accounts
Username and password can only be changed if you remember your current one to ensure security of accounts. Moreover, usernames cannot contain special characters and passwords must be at least 8 characters long.

Content Responsibility
You are responsible for the images you submit as evidence. Hence, all sensitive information must be blurred before submission or moderators can flag your account to developers for termination of account depending on the severity of the content. This is to the developers' discretion.

Conflicts & Contact
For any conflicts or issues you might have with the website, please contact us at:

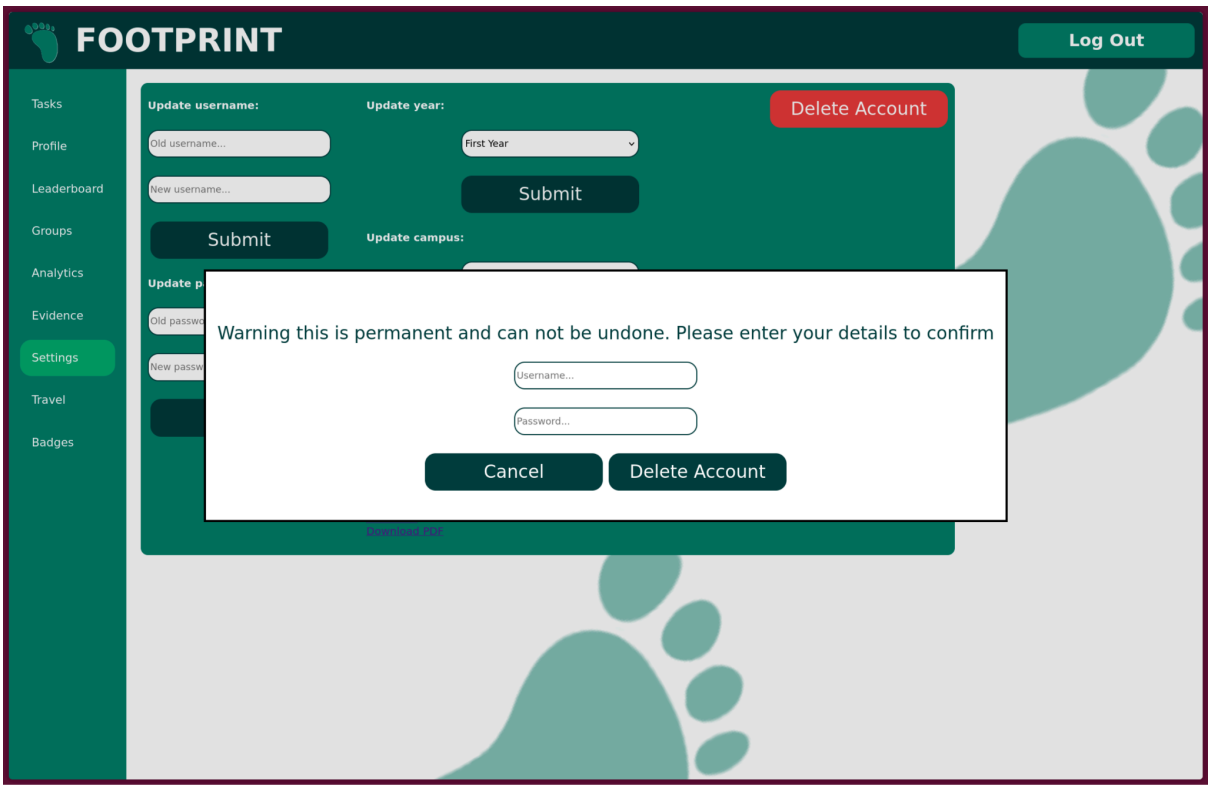
Phone Number: +44 0123456789
Email: mw1104@exeter.ac.uk or zbp201@exeter.ac.uk
Address: Innovation Centre, Rennes Drive, Exeter, EX4 4RN, England

We will do our best to resolve them within 2 weeks of contact.

Termination
You reserve the right to terminate your account at any time, provided you remember your username and password. This is done as a security measure.

Law
This website operates under UK law.

The above is demonstrated as shown below:



Ethical Considerations

- ## Fairness & Transparency [8]

  To use the website, the user must give their consent. Hence, they are aware that their personal data such as the groups they are in, will be used for analytics. Only necessary data is processed.

  A key aspect of the website is the analytics and the user is aware of that before using the website as this must be said to the users before sending them the website. They are also aware that their photo evidences are kept for a certain duration of time for the moderators to be able to check if they are submitting the same image repeatedly. This is mentioned in the terms and conditions for using the website. Moreover, moderators will have to provide a reason for approval/denial of evidence hence making the whole process transparent.

  The calculation done for the amount of carbon saved is transparent and visible to users. For each task, the amount of carbon saved is mentioned and the points earned are a direct correlation to that. For any further information needed by the users, they can always reach out to the development team.

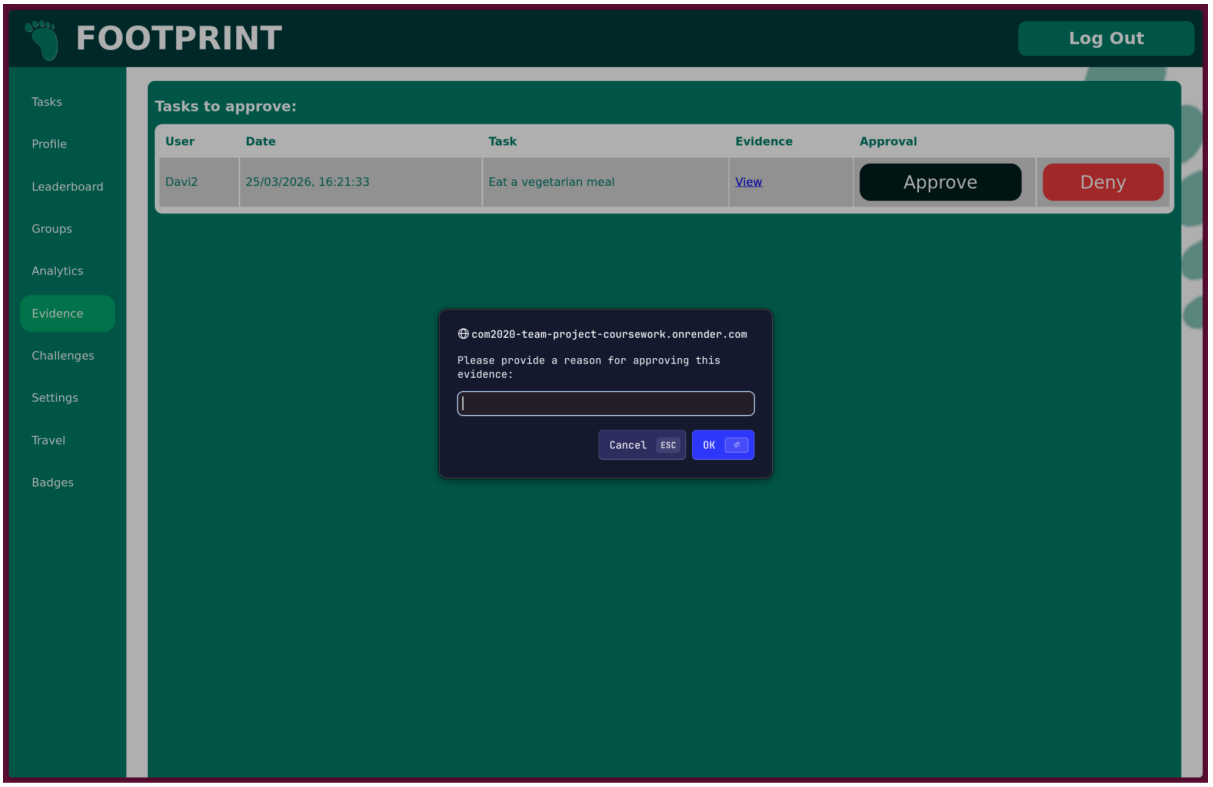
  Furthermore, users are prevented from spamming the completion of challenges by showing a pop-up telling them they have already completed the task and can only redo it after a certain time period. The travel tab also requires evidence for any travel exceeding 20km as we have assumed that 20km+ is unusual.

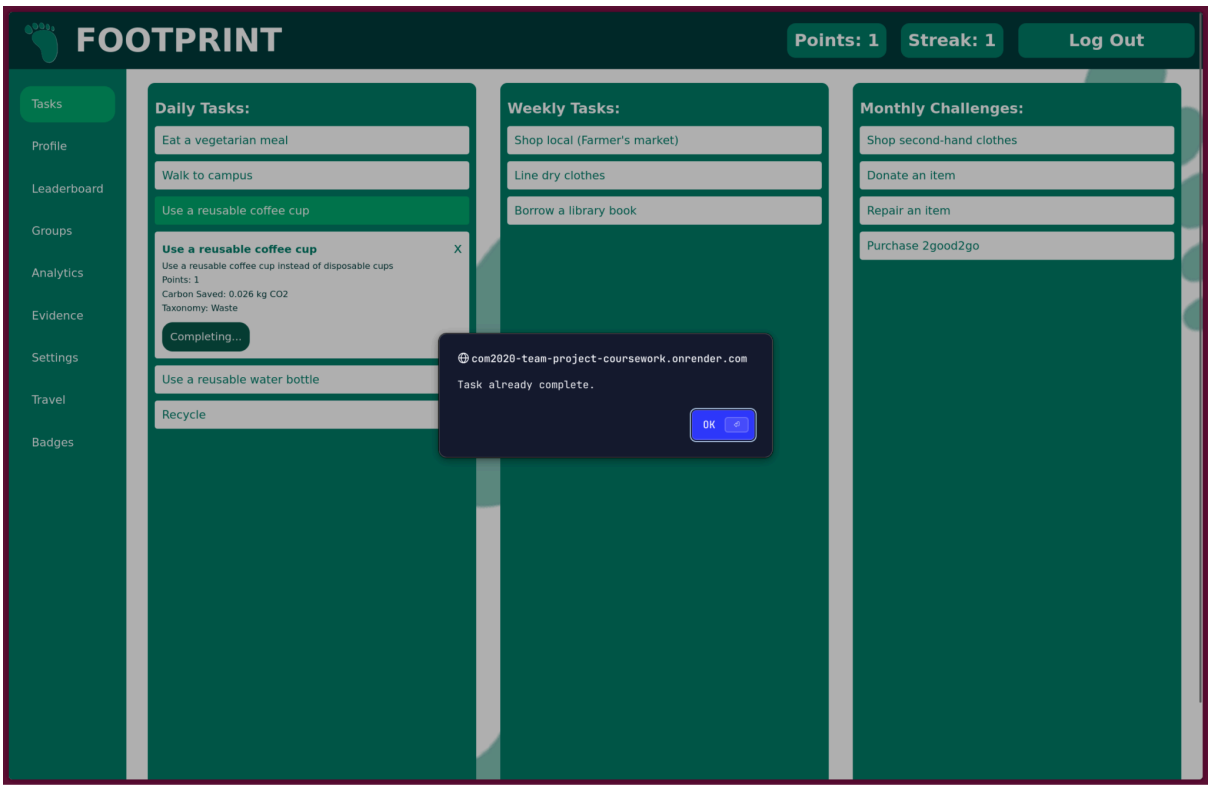
Moderators can not complete tasks requiring evidence as they would be able to approve their own evidence which is unfair. Hence, if the moderators want to participate as well, they can only do non-evidence based tasks.

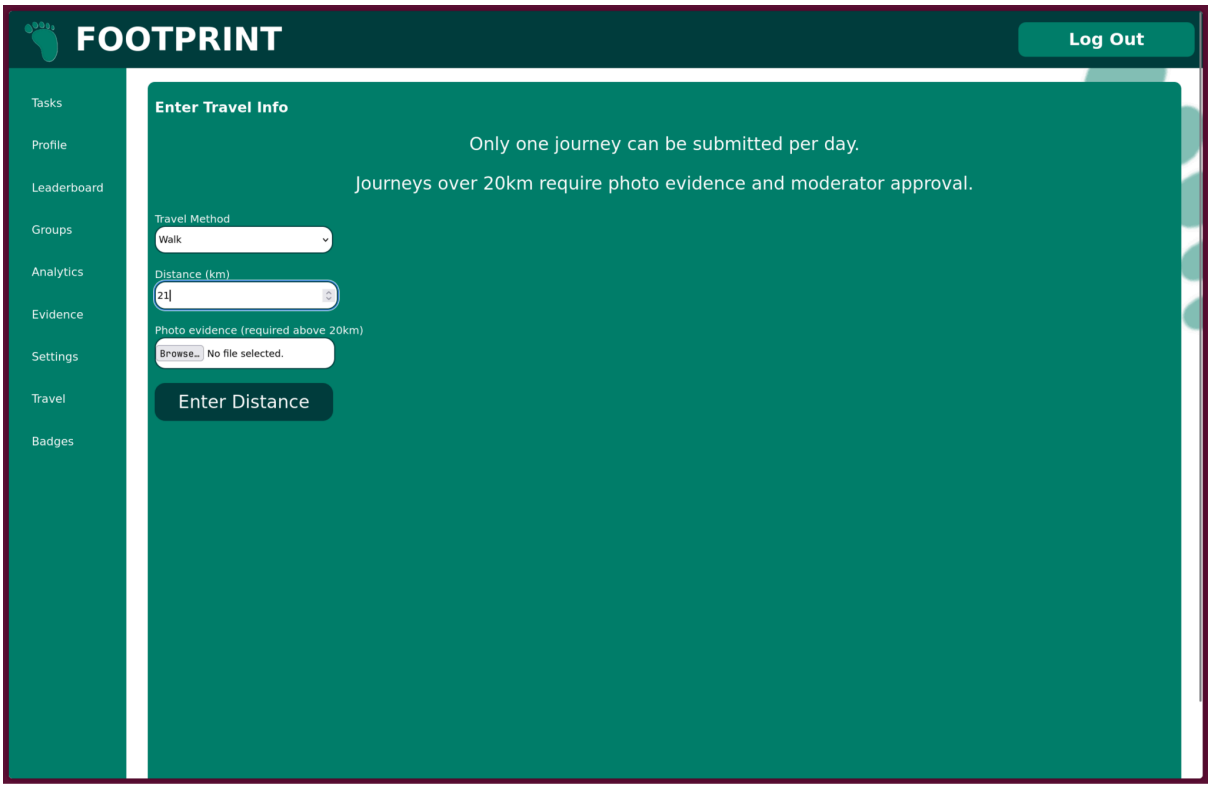
- ## Incentives

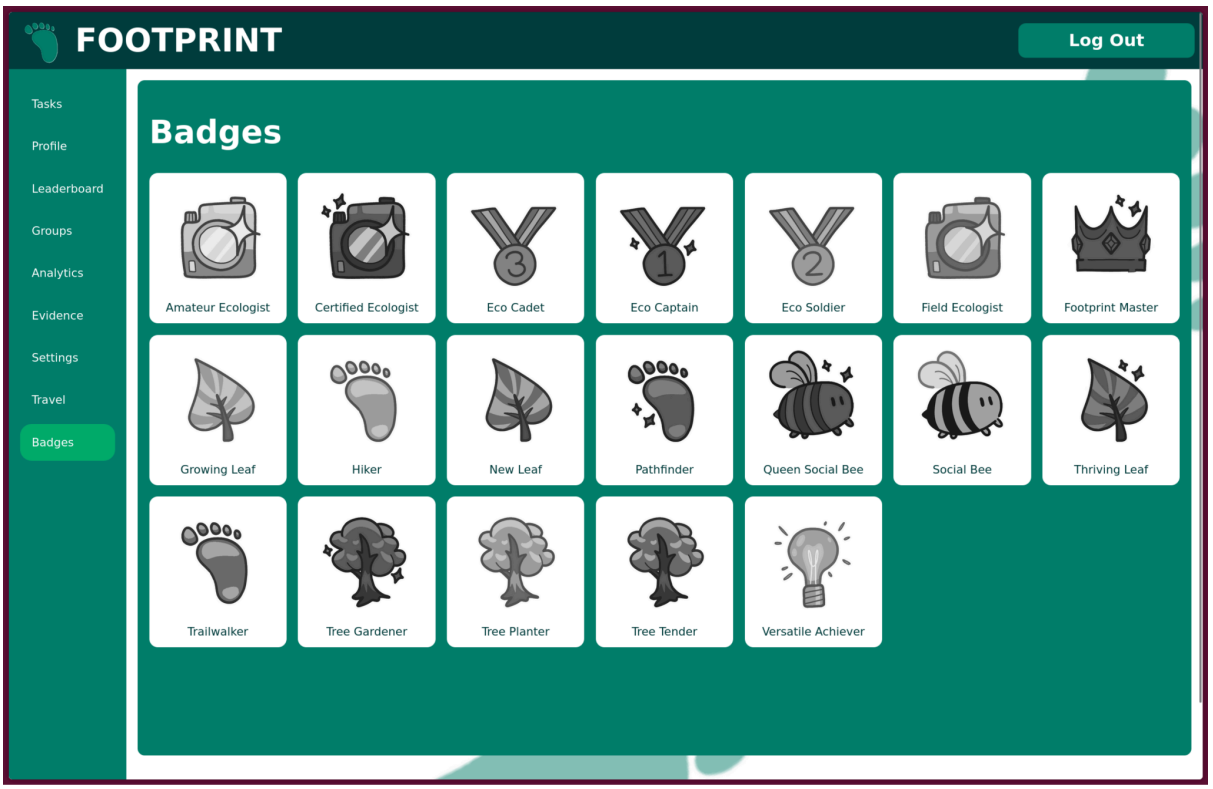
  Incentives include the gamification of the website. Badges, streaks, groups and an analytics tab were added.
  Badges and streaks drive engagement and help foster habits. Streaks primarily promote daily consistency through the fear of loss, badges serve as visual milestones that celebrate achievements and provide social proof. Additionally, being able to participate as a group encourages more people to use the website as well as introducing an aspect of positive peer pressure. Having the analytics tab to keep track of their improvements further inspire them to continuously decrease their carbon footprint and have a sense of achievement upon seeing their progress.

The above is demonstrated as shown below:

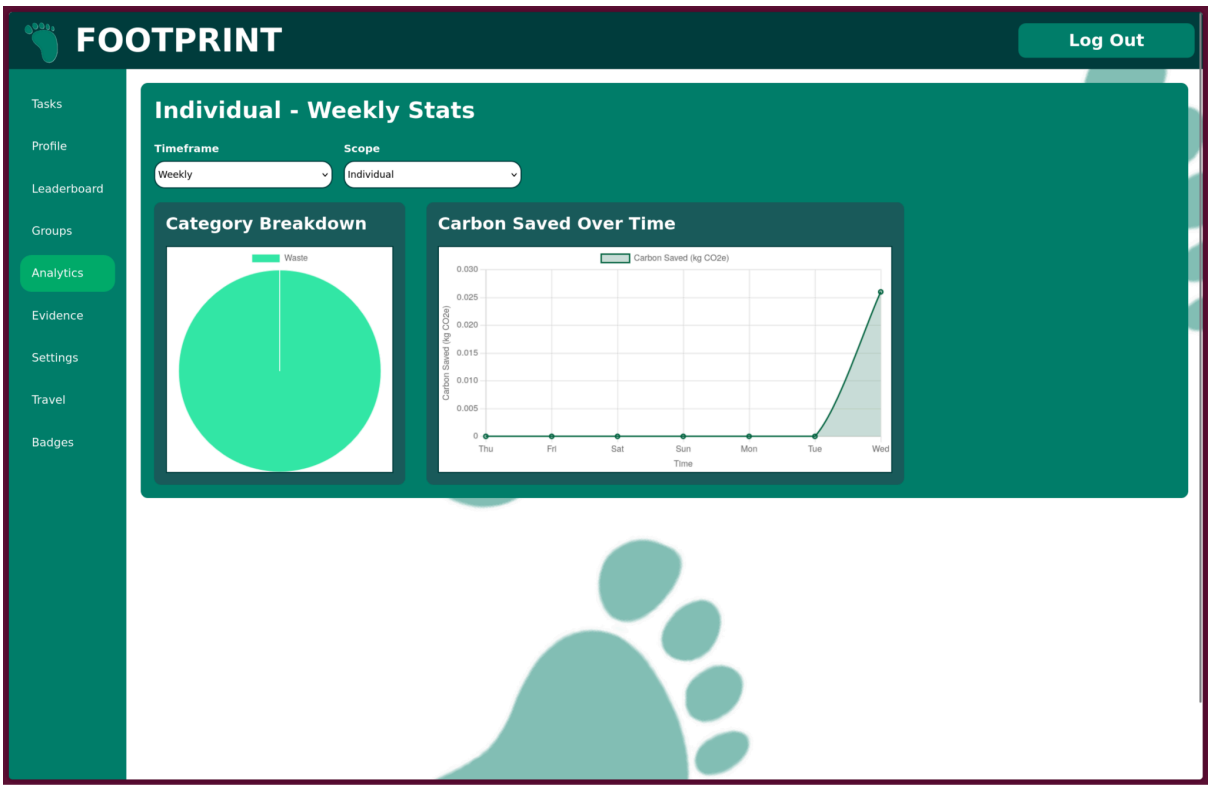


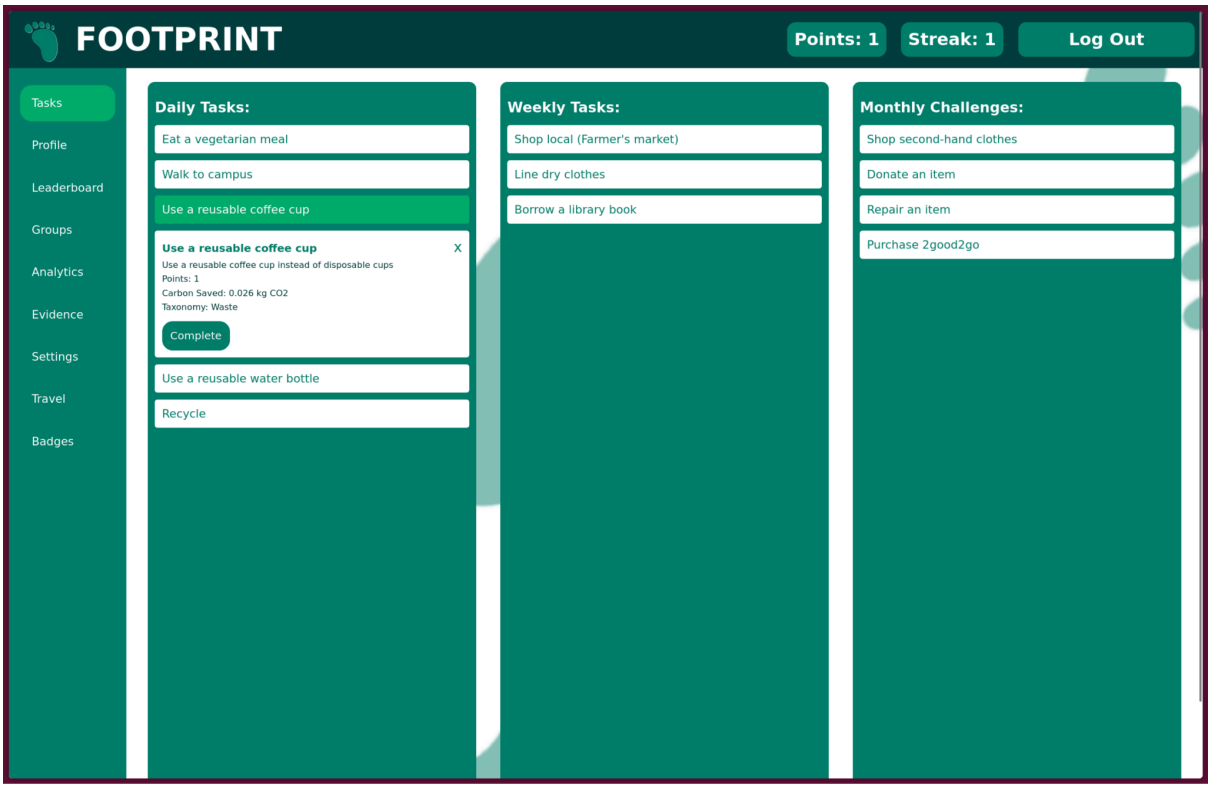

FOOTPRINT
Points: 1
Streak: 1
Log Out
Tasks
Profile
Leaderboard
Groups
Analytics
Evidence
Settings
Travel
Badges
Daily Tasks:
Eat a vegetarian meal
Walk to campus
Use a reusable coffee cup
Use a reusable coffee cup
X
Use a reusable coffee cup instead of disposable cups
Points: 1
Carbon Saved: 0.026 kg CO2
Taxonomy: Waste
Completing...
Use a reusable water bottle
Recycle
Weekly Tasks:
Shop local (Farmer's market)
Line dry clothes
Borrow a library book
Monthly Challenges:
Shop second-hand clothes
Donate an item
Repair an item
Purchase 2good2go
com2020-team-project-coursework.onrender.com
Task already complete.
OK


FOOTPRINT
Log Out
Tasks
Profile
Leaderboard
Groups
Analytics
Evidence
Settings
Travel
Badges
Enter Travel Info
Only one journey can be submitted per day.
Journeys over 20km require photo evidence and moderator approval.
Travel Method
Walk
Distance (km)
21
Photo evidence (required above 20km)
Browse... No file selected.
Enter Distance


FOOTPRINT
Log Out
Tasks
Profile
Leaderboard
Groups
Analytics
Evidence
Settings
Travel
Badges
Badges
Amateur Ecologist
Certified Ecologist
Eco Cadet
Eco Captain
Eco Soldier
Field Ecologist
Footprint Master
Growing Leaf
Hiker
New Leaf
Pathfinder
Queen Social Bee
Social Bee
Thriving Leaf
Trailwalker
Tree Gardener
Tree Planter
Tree Tender
Versatile Achiever

FOOTPRINT
Log Out
Tasks
Profile
Leaderboard
Groups
Analytics
Evidence
Settings
Travel
Badges
Join a group:
Group Code...
Join Group
Create a group:
Group Name...
Create Group
Groups:
Test Group
Points: 1
Code: 9917CEAF
Disband





References

1. *Data protection: The UK's data protection legislation - GOV.UK*
2. *Passwords in online services | ICO*

3. *Spring Security - Implementation of BCryptPasswordEncoder - GeeksforGeeks*
4. *https://share.google/iuCSUp4B64Gco7mKo*
5. *Understanding WCAG 2.2 - Service Manual - GOV.UK*
6. *Free Terms & Conditions Template & Examples [PDF+DOC]*
7. Website Legal Compliance Documents UK 2025: Complete Guide
8. *gdprlocal.com/lawfulness-fairness-and-transparency-gdpr-principles/*
9. https://www.section508.gov/manage/laws-and-policies/website-accessibility-statement/
10. https://www.gov.uk/government/publications/sample-accessibility-statement/sample-accessibility-statement-for-a-fictional-public-sector-website
11. https://www.gov.uk/government/publications/cma-data-retention-policy/data-retention-policy